

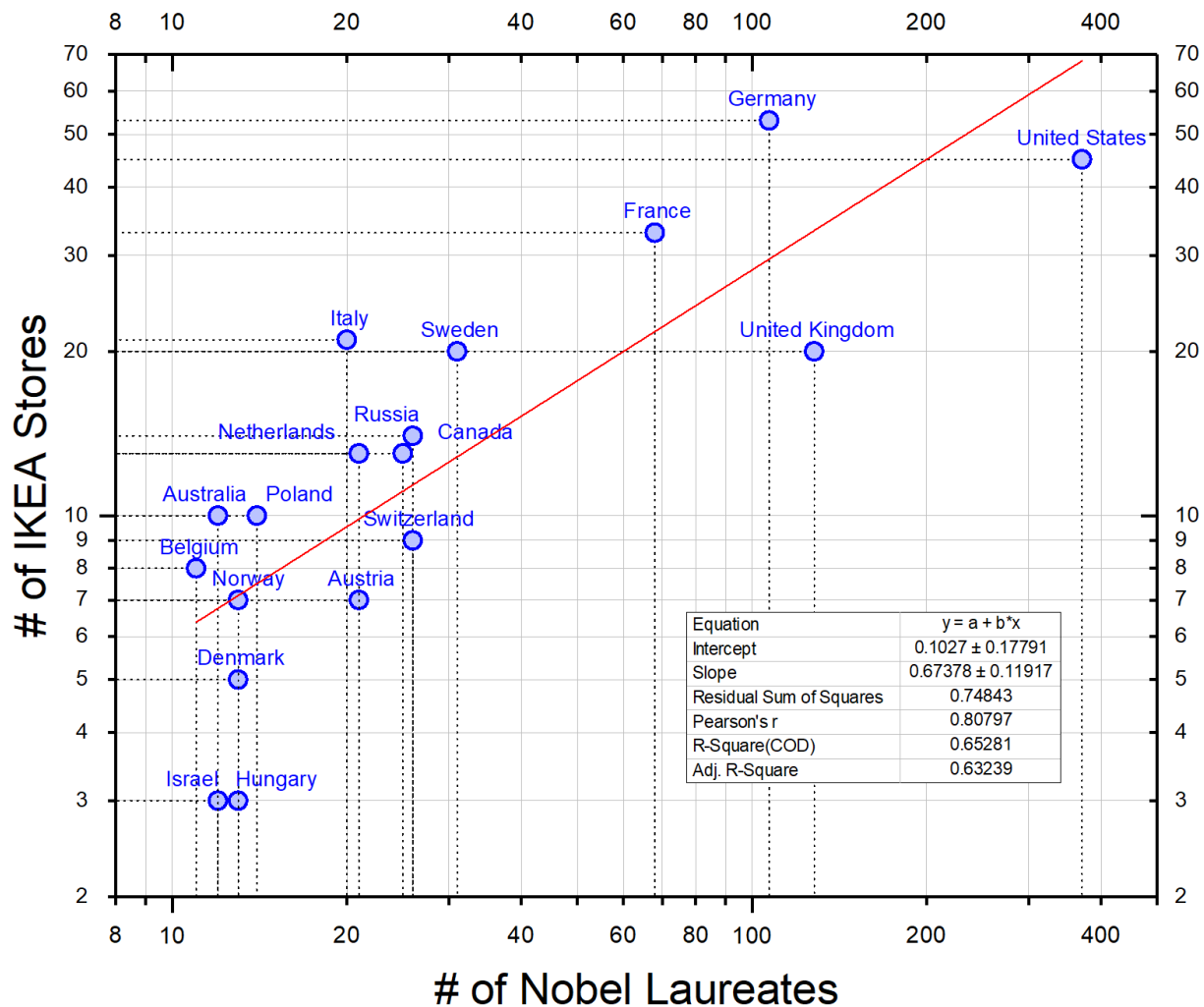
Advanced Econometrics – Exam

September 1, 2025

Exercise 1 (30%)

Your lecturer intrigued by a plot¹ and the paper by Maurage et al. (2013)² decided to check how certain factors affect the number of Nobel Prize Laureates. IKEA is a multinational conglomerate founded in Sweden that designs and sells ready-to-assemble furniture, household goods, and various related services.

Figure 1: Number of IKEA stores and Nobel Prize laureates



The dataset contains the following variables:

- **country**: country identifier (i)
- **nobel**: number of Nobel prize laureates in the i -th country
- **ln population**: logarithm of population of the i -th country
- **gdppc**: GDP per capita of the i -th country

Assume $\alpha = 5\%$ significance level. Using the following outputs answer the following questions. Table 1 presents results of the Poisson regressions (pois1, pois2), the negative binomial regressions (nb1, nb2) and the zero-inflated Poisson regression (zip).

¹<https://www.reddit.com/r/dataisbeautiful/>

²Pierre Maurage, P., Alexandre Heeren, A., Pesenti, M. (2013) Does Chocolate Consumption Really Boost Nobel Award Chances? The Peril of Over-Interpreting Correlations in Health Studies. *The Journal of Nutrition*, 143(6), 931-933. <https://doi.org/10.3945/jn.113.174813>

Table 1: Number of Nobel prize laureates and their frequency in the dataset

Nobel	0	1	2	3	4
Frequency	57.1%	14.8%	7.6%	3.2%	0.5%
Nobel	5	6	8	11	13
Frequency	1.0%	2.1%	1.0%	1.6%	1.0%
Nobel	14	15	18	21	22
Frequency	1.6%	0.5%	0.5%	0.5%	0.5%
Nobel	26	27	28	30	31
Frequency	0.5%	0.5%	0.5%	0.5%	0.5%
Nobel	34	75	115	142	420
Frequency	0.5%	0.5%	0.5%	0.5%	0.5%

- a) (5%) Poiss1 column in Table 1 contains estimates of the Poisson regression model estimated on the sample of countries which had at least 1 IKEA store. The rest of the columns presents estimates of the models estimated on the full sample of countries (all countries). In the analysis should we focus on countries with non-zero IKEA stores or entire population of countries? Explain.
- b) (5%) Do we have excess of zeros in the Nobel variable distribution? Explain.
- c) (5%) Interpret parameters of the zero-inflated Poisson regression model.
- d) (5%) Out of the models presented in Table 1, select the superior one. Explain.
- e) (5%) Comment on lrtest and explain your answer.
- f) (5%) Is the relationship between Nobel prize laureates and number of IKEA stores real or spurious? Explain.

Table 2: Models' results

	(1) pois1	(2) pois2	(3) nb1	(4) nb2	(5) zip
<i>nobel</i>					
(Intercept)	1.475***	-0.616***	0.055	-2.553***	1.084***
IKEA_stores	0.077***	0.080***	0.184***	0.173***	0.072***
ln population		0.196***		0.360***	0.067**
<i>inflate</i>					
(Intercept)					4.313***
IKEA_stores					-0.309***
ln population					-0.504***
AIC	1278.5	1852.4	646.2	615.4	1436.1
BIC	1283.2	1861.8	655.7	628.0	1455.0
Log-Likelihood	-637.2	-923.1	-320.0	-303.7	-712.0
N	61	182	182	182	182

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

```
> lrtest(POIS2, NB2)
```

```
Likelihood ratio test
```

```
Model 1: Nobel ~ Ikea_stores + e_pop
```

```
Model 2: Nobel ~ Ikea_stores + e_pop
```

```
  #Df  LogLik Df Chisq Pr(>Chisq)
```

```
1    3 -923.19
```

```
2    4 -303.70  1  1239  < 2.2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Exercise 2 (30 %)

A data scientist analyzed cat adoptions³ from an animal shelter. The database provided information on 1000 persons who decided to adopt a cat from an animal shelter. Each and every person decided to adopt one of the five cats offered. The options of the cats were the following.

- Black cat,
- Ginger (red/orange) cat,
- Tricolor (25% to 75% white with large orange and black patches) cat,
- White cat,
- Grey-brown cat.

The independent variables were:

- **person_id**: id of the person who adopts a cat
- **cat_color**: color of the cat as defined above
- **cat_gender**: dummy variable for the cat's gender (1: female, 0: male)
- **cat_age**: age of the cat in years
- **cat_sterilized**: dummy variable for neutered/sterilized⁴ cats (1: yes; 0: no)
- **person_income**: income of the adopting person in thousands zloty
- **person_superstitious**: dummy variable indicating whether the adopting person is superstitious (1: yes; 0: no)
- **person_gender**: dummy variable for the gender of the adopting person (1: female; 0: otherwise)
- **person_age**: age of the adopting person in years

Assume $\alpha = 10\%$ significance level. Using the output below answer the following questions.

- (5%) Is the **cat_color** variable ordered or unordered? Explain.
- (5%) Which variables, if any, are alternative-specific? Explain.
- (5%) Does the R output below present a multinomial or conditional logit? Explain.
- (5%) Interpret the parameter for the **cat_age** variable.
- (5%) Interpret the parameter for the **person_gender:Ginger** variable.
- (5%) Does being superstitious affect the color of the adopted cat? Explain.

³If you choose to adopt a cat instead of buying one, you'll be providing a new home for a cat in need. Source: <https://www.cats.org.uk/help-and-advice/getting-a-cat/adopting-a-cat>

⁴Neutering, sometimes called spaying for female cats or castration for males, involves an operation to prevent female cats getting pregnant and male cats from getting females pregnant. Source: <https://www.cats.org.uk/>

Call:

```
mlogit(formula = cat_choice ~ cat_sterilized + cat_gender + cat_age |
      person_age + person_superstitious + person_gender, data = df_new,
      reflevel = "White", method = "nr")
```

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)	
(Intercept):Black	-1.086232	1.322922	-0.8211	0.411598	
(Intercept):Ginger	-3.200237	1.227337	-2.6075	0.009122	**
(Intercept):Grey-brown	1.642942	1.049985	1.5647	0.117646	
(Intercept):Triplecolor	-1.251403	1.283365	-0.9751	0.329513	
cat_sterilized	1.392461	0.177141	7.8608	0.000000	***
cat_gender	-0.266623	0.191367	-1.3933	0.163542	
cat_age	-0.152705	0.056063	-2.7238	0.006454	**
person_age:Black	0.023511	0.037563	0.6259	0.531379	
person_age:Ginger	0.062907	0.032482	1.9367	0.052785	.
person_age:Grey-brown	0.051502	0.030956	1.6637	0.096170	.
person_age:Triplecolor	0.026969	0.036378	0.7414	0.458474	
person_superstitious:Black	17.290975	2347.170175	0.0074	0.994122	
person_superstitious:Ginger	17.249291	2347.170133	0.0073	0.994136	
person_superstitious:Grey-brown	17.392295	2347.170129	0.0074	0.994088	
person_superstitious:Triplecolor	17.699235	2347.170157	0.0075	0.993983	
person_gender:Black	0.189792	0.637885	0.2975	0.766059	
person_gender:Ginger	4.917490	0.643130	7.6462	0.000000	***
person_gender:Grey-brown	-3.375449	0.571623	-5.9050	0.000000	***
person_gender:Triplecolor	0.504711	0.614276	0.8216	0.411285	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -366.37

McFadden R²: 0.62178

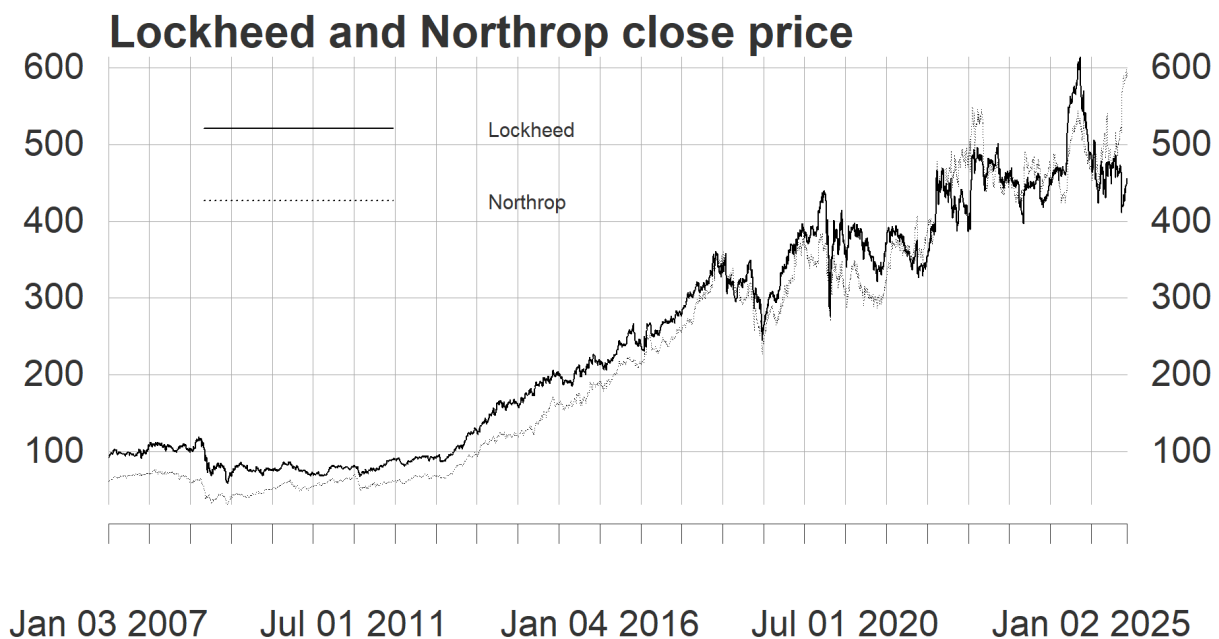
Likelihood ratio test : chisq = 1204.6 (p.value = < 0.0000000000000000222)

Exercise 3 (40%)

Lockheed Martin and Northrop Grumman are two out of five "Big Five" military corporations in the United States. These companies produce modern warfare systems, and stealth airplanes among other products. The great share of revenue of these two companies comes from the United States government.

Your lecturer hypothesized that stock prices of the Lockheed Martin and Northrop Grumman companies may be cointegrated. The data covered daily (working days) observations for the period 2007-2025 (August 28).

Figure 2: Lockheed Martin and Northrop Grumman (dotted line, right axis) close price



Use $\alpha = 5\%$ **significance level**. The critical value for cointegration is -3.67 .

- (10%) Does visual inspection prove the idea to estimate the cointegration model? Explain.
- (8%) What is the integration order of these two time series? Explain.
- (7%) Are these two variables cointegrated? Explain.
- (10%) Interpret the estimates of the error-correction mechanism.
- (5%) Is this relationship real or spurious? Explain.

```
> testdf2(variable = fighters$Lockheed, test.type = "c",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -0.7480642 0.7812316             4693      1.178736e+00
2             1 -0.7302288 0.7879004             4692      1.159752e-05
3             2 -0.7253643 0.7897193             4691      7.942945e-06
4             3 -0.7059747 0.7969692             4690      1.076323e-03
5             4 -0.6603741 0.8140195             4689      1.705413e-02
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1      1.2261160397      1.868801459      9.79905875      0.2776134
2      0.0532409678      0.764896664      8.34466196      0.9972828
3      0.0004582499      0.701349324      8.28989581      0.9977513
4      0.0018085147      0.003593189      7.48760968      0.9738282
5      0.0172590590      0.031231344      0.03213799      0.8960985
  p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1      0.5416918      0.6000786      0.04395202
2      0.9737307      0.8578435      0.07973812
3      0.9997709      0.8728865      0.08151790
4      0.9990962      0.9999428      0.11225690
5      0.9914076      0.9985457      0.99987227
```

```
> testdf2(variable = diff.xts(fighters$Lockheed), test.type = "nc",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -69.57331 0.01             4692      7.397464e-04
2             1 -48.94580 0.01             4691      7.220273e-04
3             2 -40.26855 0.01             4690      3.225483e-03
4             3 -36.14881 0.01             4689      1.081783e-02
5             4 -30.50238 0.01             4688      5.896729e-07
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1      0.059862615      0.792099220      8.44807337      0.9783016
2      0.002715555      0.724253252      8.38766616      0.9785630
3      0.003231297      0.007604962      7.56237535      0.9547098
4      0.010966347      0.019361800      0.01937959      0.9171624
5      0.031411996      0.051254347      0.05369698      0.9993873
  p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1      0.9705122      0.8513558      0.07647692
2      0.9986431      0.8674851      0.07836631
3      0.9983857      0.9998240      0.10899005
4      0.9945318      0.9992876      0.99995336
5      0.9844167      0.9969609      0.99964597
```

```
> pp.test <- ur.pp(fighters$Northrop, type = c("Z-tau"), model = c("constant"))
> summary(pp.test)
```

```
#####
# Phillips-Perron Unit Root Test #
#####
```

Value of test-statistic, type: Z-tau is: 0.3319

```
      aux. Z statistics
Z-tau-mu      0.8466
```

```
Critical values for Z statistics:
      1pct      5pct      10pct
critical values -3.43478 -2.862684 -2.567407
```

```
> kpss.test <- ur.kpss(diff.xts(fighters$Northrop), type = c("mu"))
> summary(kpss.test)
```

```
#####
# KPSS Unit Root Test #
#####
```

Test is of type: mu with 10 lags.

Value of test-statistic is: 0.1569

```
Critical value for a significance level of:
      10pct  5pct 2.5pct  1pct
critical values 0.347 0.463 0.574 0.739
```

```
> model <- lm(fighters$Lockheed~fighters$Northrop, data=fighters)
> summary(model)
```

Call:

```
lm(formula = fighters$Lockheed ~ fighters$Northrop, data = fighters)
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  40.788349   0.610715   66.79   <2e-16 ***
fighters$Northrop  0.909216   0.002194  414.34   <2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 24.1 on 4692 degrees of freedom

Multiple R-squared: 0.9734, Adjusted R-squared: 0.9734

F-statistic: 1.717e+05 on 1 and 4692 DF, p-value: < 2.2e-16

```
> testdf2(variable = residuals(model), test.type="nc",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -3.313189  0.01             4693      1.56374662349
2             1 -3.190216  0.01             4692      0.00312496966
3             2 -2.931157  0.01             4691      0.00430549244
4             3 -3.067203  0.01             4690      0.00003157788
5             4 -3.021400  0.01             4689      0.00009614932
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1             9.285941724             12.36700531             12.46861619      0.2111175
2             7.685161143             10.83979197             10.94776829      0.9554203
3             0.004412035              3.16306489              3.33511375      0.9476834
4             0.007210950              0.02046769              0.13219198      0.9955164
5             0.005541334              0.01951947              0.02060511      0.9921764
  p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1      0.009629049      0.006226091      0.01418637
2      0.021438207      0.012624488      0.02715734
3      0.997796414      0.367160629      0.50338809
4      0.996401017      0.999225971      0.99790957
5      0.997233168      0.999278927      0.99994729

> model.ecm <- lm(diff.xts(fighters$Lockheed)~diff.xts(fighters$Northrop)+lresid-1)
> summary(model.ecm)
```

Call:

```
lm(formula = diff.xts(fighters$Lockheed) ~ diff.xts(fighters$Northrop) +
    lresid - 1)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
diff.xts(fighters\$Northrop)	0.652729	0.010000	65.271	< 2e-16 ***
lresid	-0.006407	0.001847	-3.468	0.000529 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.042 on 4691 degrees of freedom

Multiple R-squared: 0.4766, Adjusted R-squared: 0.4764

F-statistic: 2136 on 2 and 4691 DF, p-value: < 2.2e-16

